Machine learning of high-frequency trading report

Kaicheng Ni

1. Methodology

This machine learning project uses the yfinance package to capture data from Yahoo Finance. Directly delete rows with missing values when preprocessing raw data. Calculate technical indicators including short-term moving averages, RSI, and MACD. Train the model using three methods: ridge regression, LSTM network, and random forest, using technical indicators and preprocessed raw datasets. Finally, the best model was selected through MSE. The predicted result is the percentage change in the next minute's close price.

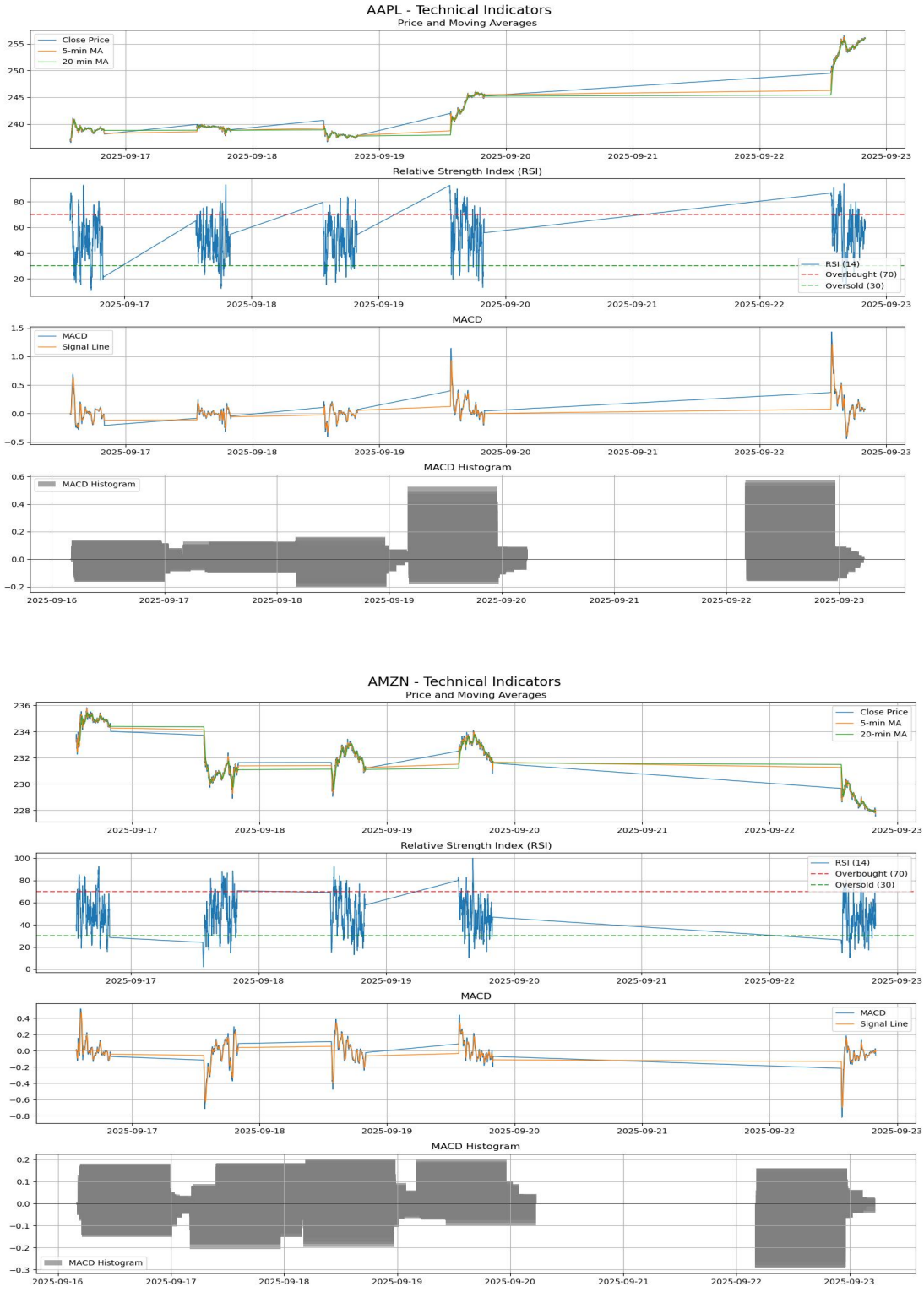
2. Data Sources

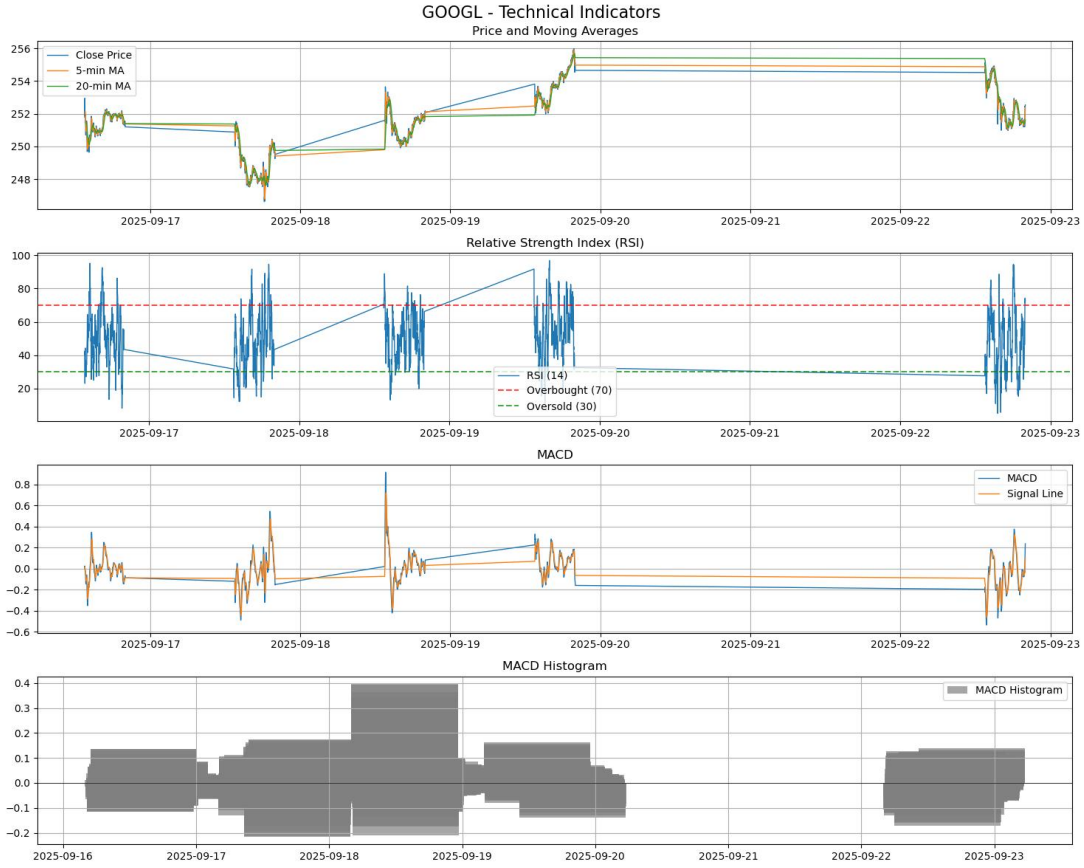
This project uses the yfinance package to extract data from Yahoo Finance, including AAPL, MSFT, GOOGL, AMZN, and NVDA for the past 5 trading days, which are the data from the past seven days. The data extracted from Yahoo Finance has a time interval of 1 minute and includes open, high, low, close, volume, and timestamp data. After completing the download, delete the missing rows using. dropna() and add a ticket column for subsequent calculations and model training.

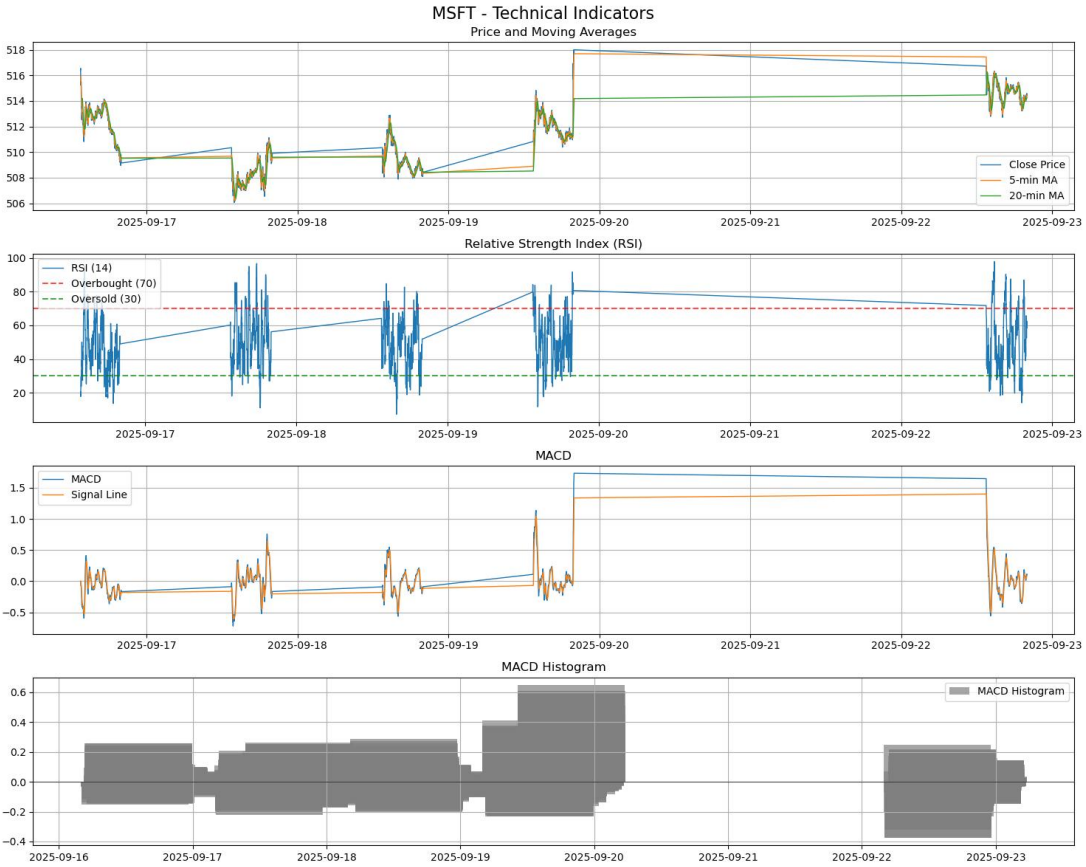
3. Feature Engineering Process

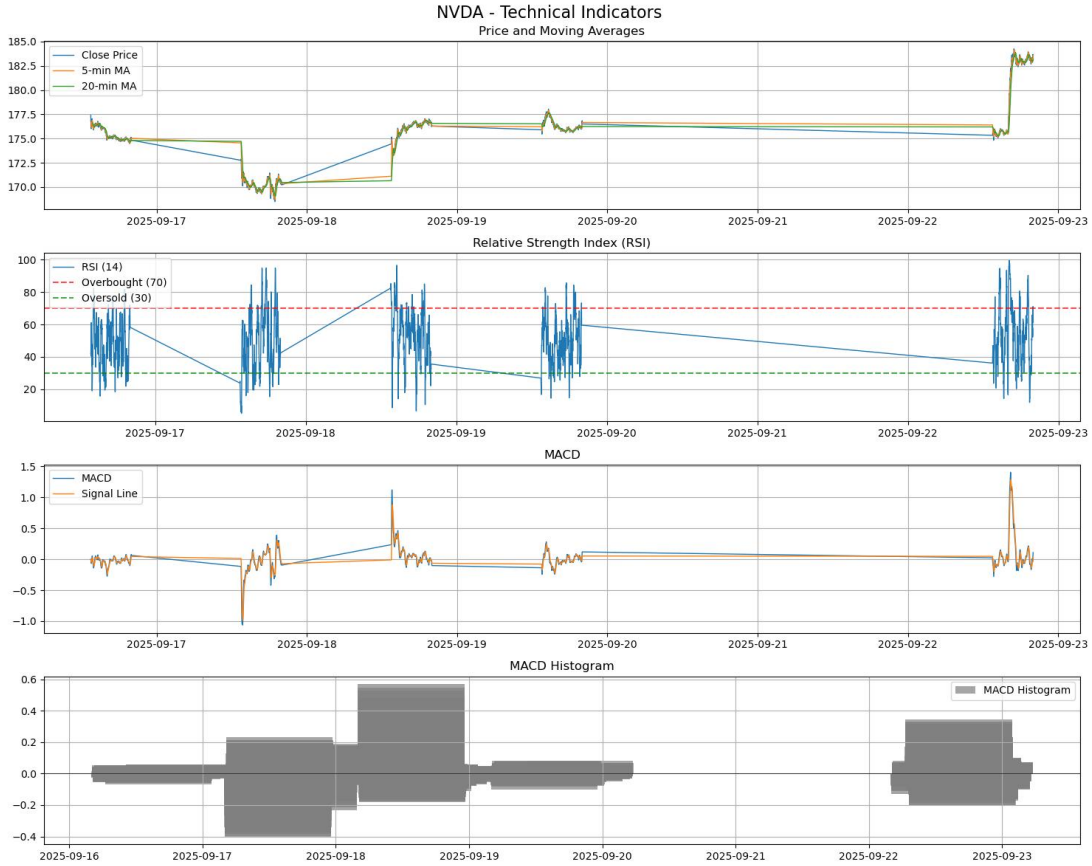
In the feature engineering process, I obtained the 5-minute and 20 minute short-term moving averages by calculating the average of the closing price sequence over the past 5 and 20 minutes. And I Obtain RSI by calculating the relative strength of the average increase and decrease in the closing price within 14 minutes, and normalizing it to a value of 0-100. This measures the momentum state of the price. And I obtain MACD by calculating the difference between the 12 minutes and 26 minutes

exponential moving averages. And the MACD histogram is obtained by subtracting the signal line from the MACD line, where the signal line is the 9 minutes exponential moving average of the MACD line, visually representing the momentum gap between the two lines.These variables are also stored together with the original data in the “all_stocks_with_indicators.csv” file. The results obtained by extracting features from five tickets are shown in the following figure:




GOOGL - Technical Indicators
Price and Moving Averages
Close Price
5-min MA
20-min MA
Relative Strength Index (RSI)
RSI (14)
Overbought (70)
Oversold (30)
MACD
MACD
Signal Line
MACD Histogram
MACD Histogram


MSFT - Technical Indicators
Price and Moving Averages
Close Price
5-min MA
20-min MA
Relative Strength Index (RSI)
RSI (14)
Overbought (70)
Oversold (30)
MACD
MACD
Signal Line
MACD Histogram
MACD Histogram



4. Model Architecture And Training Process

This training process implements a multi-tier price prediction architecture that begins with standardizing raw feature data through normalization, then concurrently trains three distinct types of forecasting models. For linear modeling, it employs Ridge regression with L2 regularization, optimizing the regularization strength through grid search. On the ensemble learning front, it utilizes a random forest comprising 100 decision trees, enhancing generalization capability via Bagging strategy. For deep learning, it constructs a dual-layer LSTM network where the first layer with 50 units preserves sequence outputs, the second layer with 50 units generates final states, incorporates 20% dropout layers between them to prevent overfitting, and ultimately outputs predictions through fully connected layers. During the evaluation phase, the system comprehensively compares four key metrics: mean squared error, mean absolute error, R-squared coefficient, and directional accuracy to select the optimal model for each stock based on the principle of minimizing mean squared error, thereby forming a complete prediction pipeline spanning from feature processing to

model optimization.

5. Evaluation Results

Evaluation results are as follows:

MSFT Model Performance:
Ridge Regression - MSE: 0.001479, MAE: 0.027996, R^2: -0.020928, Direction Accuracy: 0.5233
Random Forest - MSE: 0.004103, MAE: 0.053904, R^2: -1.832193, Direction Accuracy: 0.5104
LSTM - MSE: 0.001362, MAE: 0.027655, R^2: -0.051745, Direction Accuracy: 0.5160
MSFT Best Model: LSTM

AAPL Model Performance:
Ridge Regression - MSE: 0.005048, MAE: 0.051256, R^2: -0.115141, Direction Accuracy: 0.5466
Random Forest - MSE: 0.007305, MAE: 0.060565, R^2: -0.613686, Direction Accuracy: 0.4352
LSTM - MSE: 0.004333, MAE: 0.048788, R^2: -0.072717, Direction Accuracy: 0.5266
AAPL Best Model: LSTM

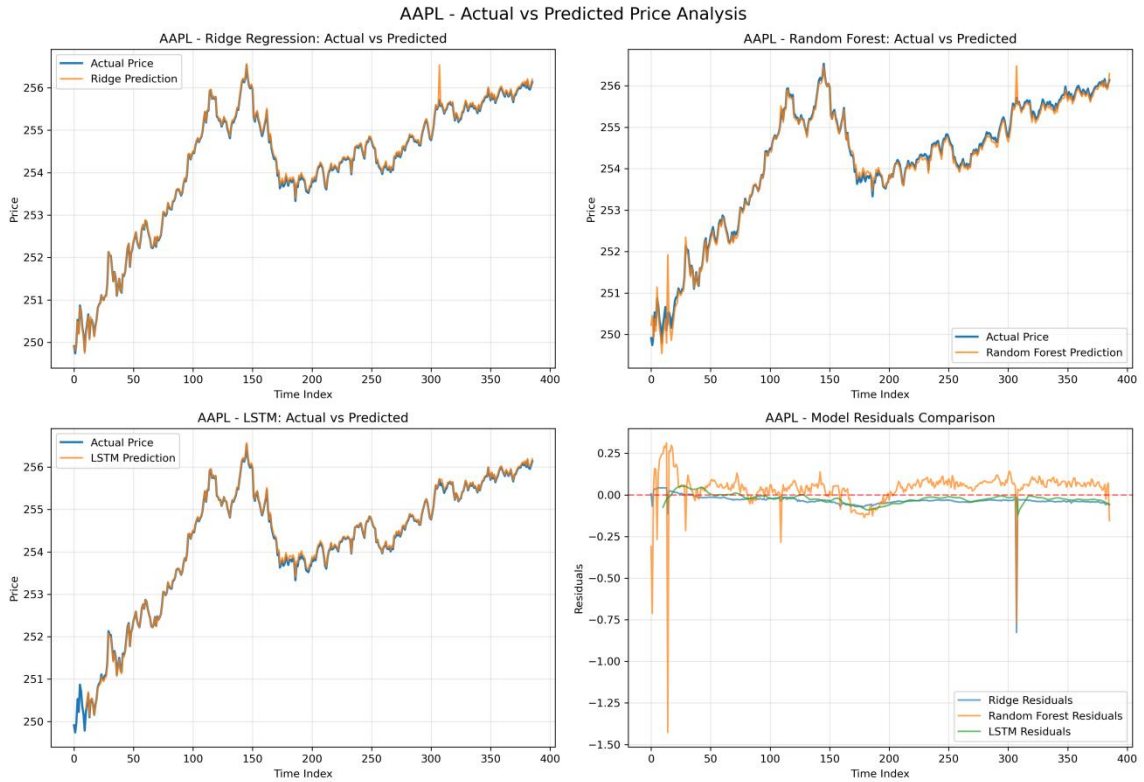
GOOGL Model Performance:
Ridge Regression - MSE: 0.004946, MAE: 0.042760, R^2: -0.049352, Direction Accuracy: 0.4715
Random Forest - MSE: 0.004682, MAE: 0.042794, R^2: 0.006739, Direction Accuracy: 0.5389
LSTM - MSE: 0.004555, MAE: 0.040857, R^2: -0.010468, Direction Accuracy: 0.5293
GOOGL Best Model: LSTM

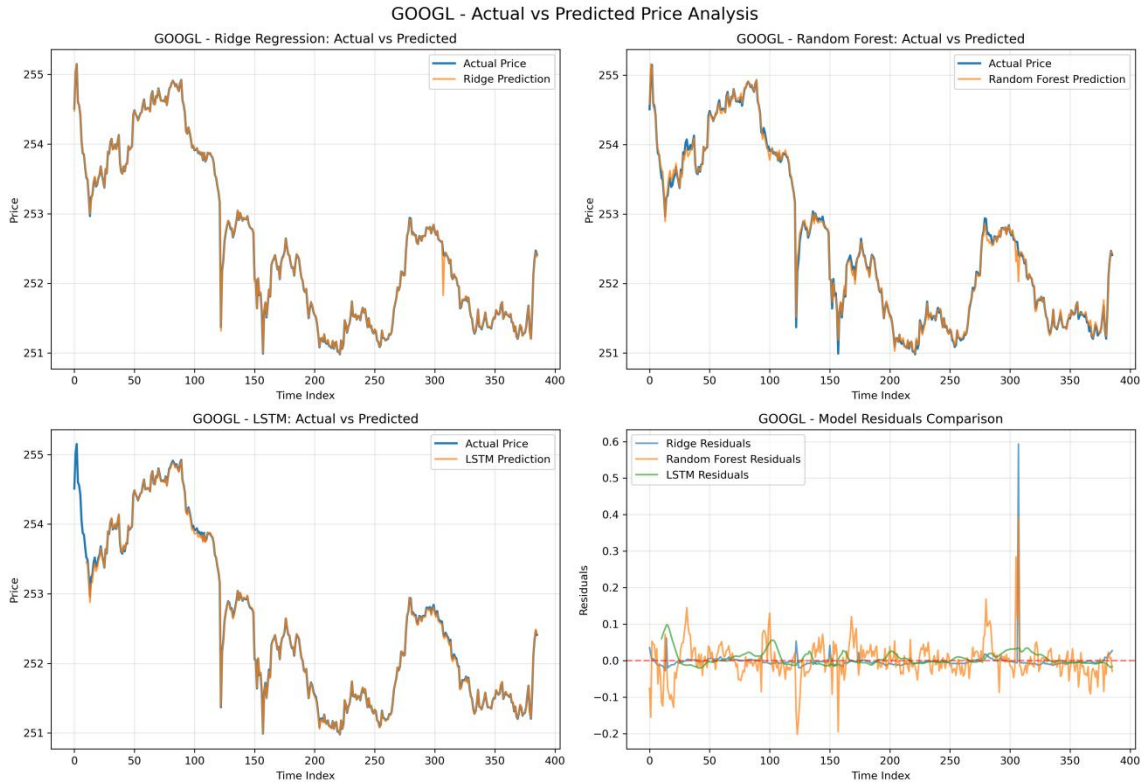
AMZN Model Performance:
Ridge Regression - MSE: 0.001709, MAE: 0.030581, R^2: -0.062201, Direction Accuracy: 0.4974
Random Forest - MSE: 0.009671, MAE: 0.088778, R^2: -5.010893, Direction Accuracy: 0.4948
LSTM - MSE: 0.001748, MAE: 0.032279, R^2: -0.237436, Direction Accuracy: 0.5027
AMZN Best Model: Ridge

NVDA Model Performance:
Ridge Regression - MSE: 0.011166, MAE: 0.068344, R^2: -0.255834, Direction Accuracy: 0.5026
Random Forest - MSE: 0.025190, MAE: 0.103083, R^2: -1.833049, Direction

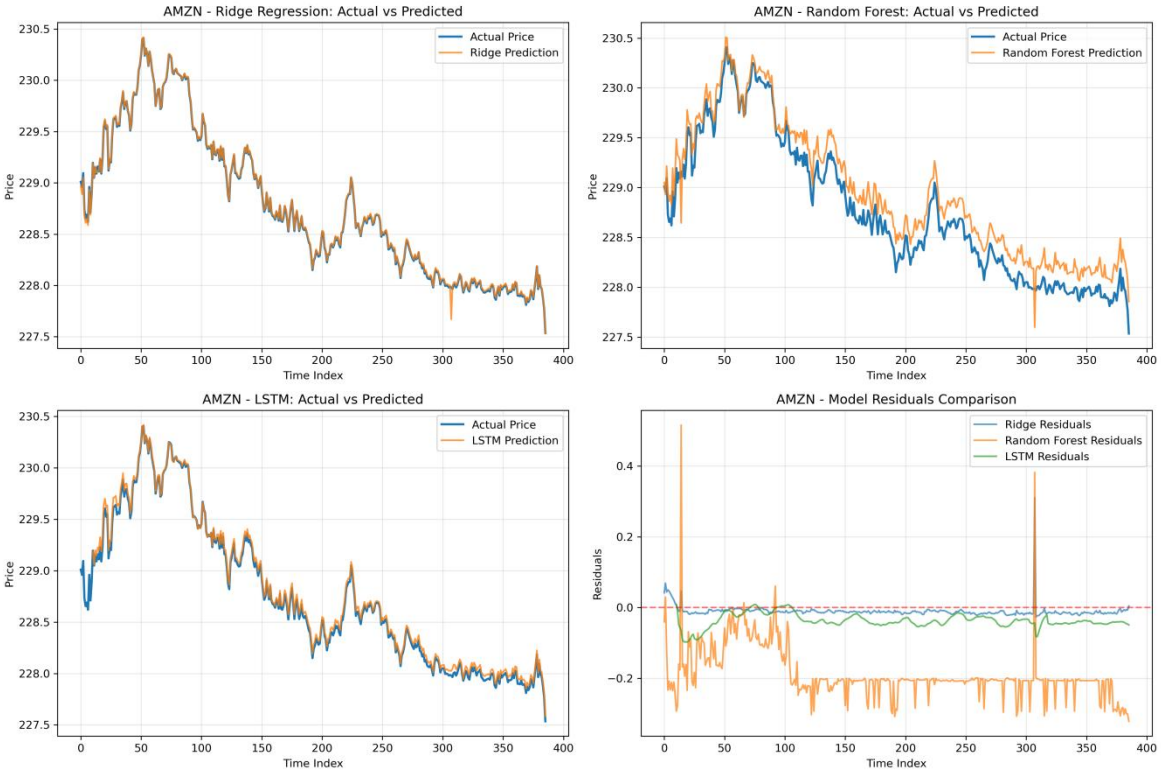
Accuracy: 0.5130
LSTM - MSE: 0.009842, MAE: 0.065868, R^2: -0.153723, Direction Accuracy: 0.4947
NVDA Best Model: LSTM

Actual VS Predicted price are as follows:

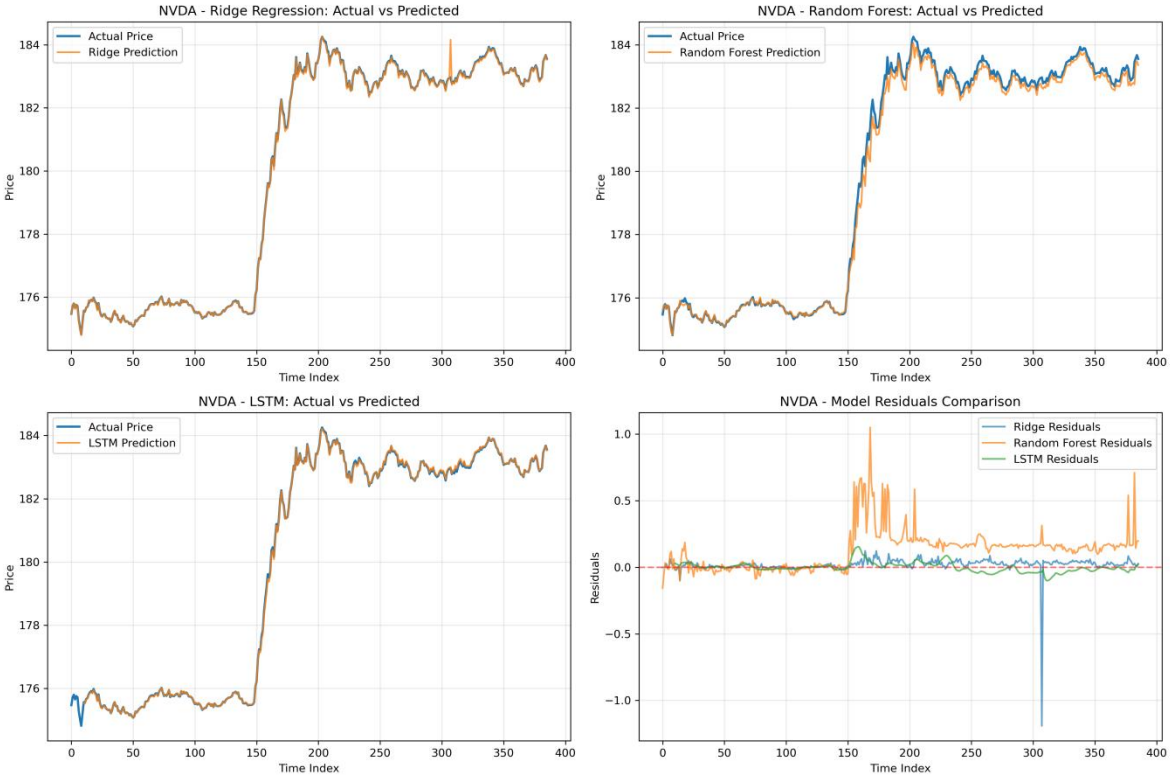


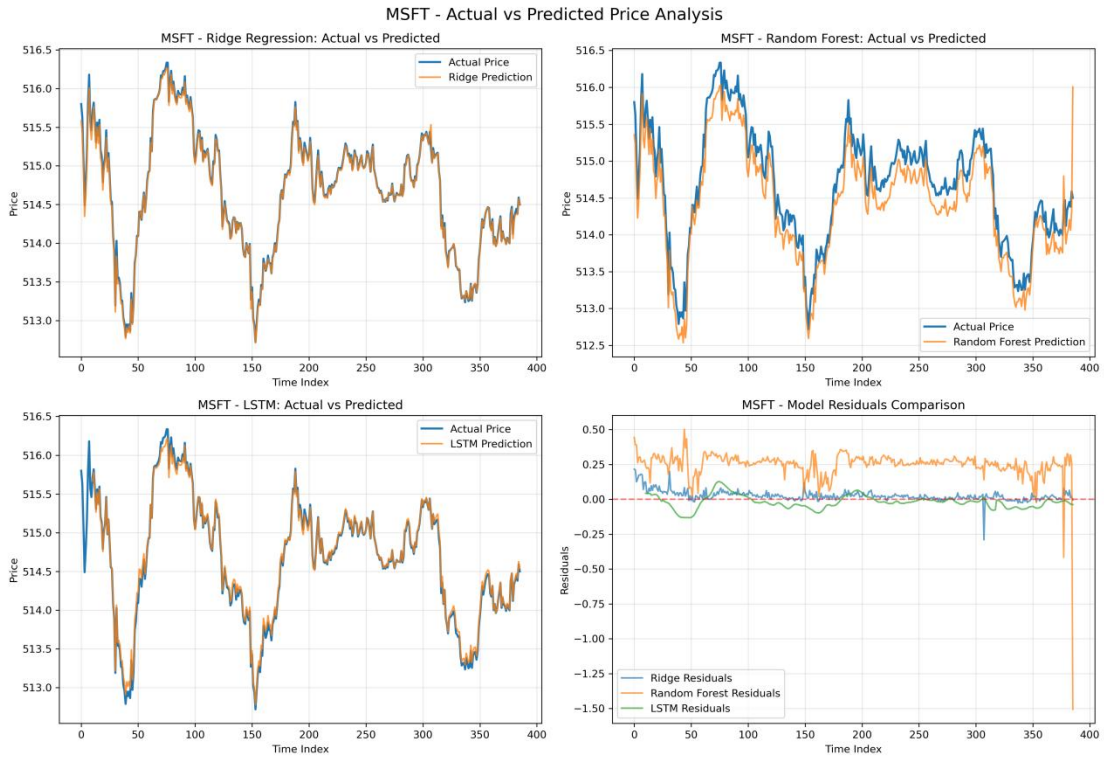


AMZN - Actual vs Predicted Price Analysis



NVDA - Actual vs Predicted Price Analysis





6. Conclusions And Recommendations

Combine the degree of agreement between the predicted results and actual results of each model on each stock, and determine the model with the best MSE performance. I think LSTM is the best model trained. I actually assigned a model training session and repeated the entire process several times. The results obtained from random forest did not yield good results in multiple training sessions at different time periods. Ridge showed better mse results than LSTM in several training sessions, but outliers were easily observed in the predicted and actual results.